

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	<i>Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.</i>		
Student Name:	Shahana	Reg. No.:	192524479
Section / Batch:		Date:	10/07/2026

Code of Conduct

I Shahana(192524479)_____

(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 -6	20
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20
GRAND TOTAL:			100 Marks

QUESTIONS
Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q2 C Code – Linear Search

10 marks

```
#include <stdio.h>
int main() {
    int arr[] = {2,4,6,8,10};
    int key = 8;
    int i;
    for(i=0;i<5;i++){
        if(arr[i]==key){
            printf("%d",i);
            break;
        }
    }
```

Your Answer

3

CO1_AT1_C CODE OUTPUT PREDICTION

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QUESTIONS
Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q3 C Code - Binary Search Complexity

10 marks

```
while(low<=high){
    mid=(low+high)/2;
}
```

Your Answer

O(log n)

QUESTIONS
Q1 ✓
C Code – Array Traversal
10 marks

Q2 ✓
C Code – Linear Search
10 marks

Q3 ✓
C Code – Binary Search Mid
10 marks

Q4 ✓
C Code – Pointer with Array
10 marks

Q5 ✓
C Code – Array Update
10 marks

Q6 ✓

Q2 C Code – Linear Search

10 marks

```
#include <stdio.h>
int main() {
    int arr[] = {2,4,6,8,10};
    int key = 8;
    int i;
    for(i=0;i<5;i++){
        if(arr[i]==key){
            printf("%d",i);
            break;
        }
    }
}
```

Your Answer

3

QUESTIONS
Q1 ✓
C Code – Array Traversal
10 marks

Q2 ✓
C Code – Linear Search
10 marks

Q3 ✓
C Code – Binary Search Mid
10 marks

Q4 ✓
C Code – Pointer with Array
10 marks

Q5 ✓
C Code – Array Update
10 marks

Q6 ✓

Q3 C Code – Binary Search Mid

10 marks

```
#include<stdio.h>
int main(){
    int low=0,high=7;
    int mid;
    mid=(low+high)/2;
    printf("%d",mid);
}
```

Your Answer

3

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QUESTIONS

Q1 C Code – Array Traversal 10 marks ✓

Q2 C Code – Linear Search 10 marks ✓

Q3 C Code – Binary Search Mid 10 marks ✓

Q4 C Code – Pointer with Array 10 marks ✓

Q5 C Code – Array Update 10 marks ✓

Q6 ✓

Q4 C Code – Pointer with Array 10 marks

```
#include<stdio.h>
int main(){
int arr[]={10,20,30};
int *p=arr;
printf("%d",*(p+2));
}
```

Your Answer

30

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QUESTIONS

Q1 C Code – Array Traversal 10 marks ✓

Q2 C Code – Linear Search 10 marks ✓

Q3 C Code – Binary Search Mid 10 marks ✓

Q4 C Code – Pointer with Array 10 marks ✓

Q5 C Code – Array Update 10 marks ✓

Q6 ✓

Q5 C Code – Array Update 10 marks

```
#include<stdio.h>
int main(){
int arr[]={1,2,3};
arr[1]=arr[0]+arr[2];
printf("%d",arr[1]);
}
```

Your Answer

4

Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q6
C Code – Linked List Node

Q6 C Code - Linked List Node

10 marks

```
#include<stdio.h>
struct Node{
int data;
struct Node *next;
};
int main(){
struct Node n1,n2;
n1.data=10;
n2.data=20;
n1.next=&n2;
n2.next=NULL;
printf("%d",n1.next->data);
}
```

Your Answer

20

QUESTIONS

Q1
C Code – Array Traversal
10 marks

Q2
C Code – Linear Search
10 marks

Q3
C Code – Binary Search Mid
10 marks

Q4
C Code – Pointer with Array
10 marks

Q5
C Code – Array Update
10 marks

Q7 C Code - Time Complexity Loop

10 marks

```
for(i=0;i<n;i++)
printf("%*");
The complexity is
```

Your Answer

O(n)

 Save Answer

CO1_AT1_C CODE OUTPUT PREDICTION

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QUESTIONS

- Q1**
C Code – Array Traversal
10 marks
- Q2**
C Code – Linear Search
10 marks
- Q3**
C Code – Binary Search Mid
10 marks
- Q4**
C Code – Pointer with Array
10 marks
- Q5**
C Code – Array Update

Q8 C Code - Nested Loop

10 marks

```
for(i=0;i<n;i++)
for(j=0;j<n;j++)
printf(" ");
```

Your Answer

O(n²)

CO1_AT1_C CODE OUTPUT PREDICTION

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QUESTIONS

- Q1**
C Code – Array Traversal
10 marks
- Q2**
C Code – Linear Search
10 marks
- Q3**
C Code – Binary Search Mid
10 marks
- Q4**
C Code – Pointer with Array
10 marks
- Q5**
C Code – Array Update

Q10 C Code - Pointer Increment

10 marks

```
#include<stdio.h>
int main(){
int a[]={5,10,15};
int *p=a;
p++;
printf("%d",*p);
}
```

Your Answer

10